

CS2011 Foundations of Computing Systems

Test 1

Date: 10th September**Duration:** 2 hours**Maximum marks:** 50

1. Consider the Boolean function given below, where $+$ denotes OR and $\cdot$ denotes AND.

$$f(x, y, z) = (x + \bar{y}) \cdot z,$$

- (a) Complete the truth table.

[2 marks]

x	y	z	f	x	y	z	f
0	0	0		1	0	0	
0	0	1		1	0	1	
0	1	0		1	1	0	
0	1	1		1	1	1	

- (b) Write the *canonical sum-of-products expression* equivalent to f .

[2 marks]

2. Simplify each expression as far as possible. Justify the important steps using Boolean laws.

(a) $(x + y)(x + \bar{y})$

(b) $xy + x(y + z)$

[4 marks]

3. A multiplexor has inputs a , b , and sel . It outputs a when $sel = 0$ and b when $sel = 1$. Write its Boolean expression and draw its logic circuit using only AND, OR, and NOT gates. **[2 marks]**

4. For each 8-bit numbers, give both its unsigned value and its signed value.

[2 marks]

00110110

11110110

5. Represent -37 in 8-bit two's complement. You need to mention the steps.

[3 marks]

6. Let x and y be two 8-bit two's-complement numbers with x positive and y negative. Argue why an adder on given the inputs x and y computes the signed sum of x and y correctly. **[4 marks]**

7. State the range of signed integers represented by an n -bit two's-complement word.

[1 marks]

8. A one-bit register initially stores 0. During each numbered cycle, the listed **in** and **load** values are applied. Write the output of the register for each cycle. **[3 marks]**

Cycle	in	load	Output
1	1	0	
2	1	1	
3	0	0	
4	0	1	
5	1	0	

9. A RAM64 is built from eight RAM8 chips. For the six-bit address 101110_2 , identify the selected RAM8 bank and the selected register within that bank. **[2 marks]**

10. Construct RAM8 using 16-bit registers and the routing chips DMux8Way and Mux8Way16. Label the shared **in**[16], **load**, **address**[3], and **out**[16] signals. **[3 marks]**

11. A Program Counter currently stores 40. Give the value stored after the next clock boundary in each independent case. The control priority is **reset**, then **load**, then **inc**, then **hold**. [2 marks]

Case	in	load	inc	reset	Next value
(i)	75	0	1	0	
(ii)	75	1	1	0	
(iii)	75	1	1	1	
(iv)	75	0	0	0	

12. Answer the following questions regarding the Hack assembly-language program given below.

```

@R1
D=M
@END
D;JLE
@R0
M=D
(END)
@END
0;JMP

```

- (a) Explain what the program does. State the final value of R0 when the initial value of R1 is (i) -3 and (ii) 7 . In each case, assume that R0 initially contains 12. [4 marks]
- (b) What does the final two instructions do? Why is the computation field 0? [2 marks]
13. Give the 16-bit binary encoding of the following instructions. [2 marks]

@21 D=D+M

Recall that a C-instruction begins with 111 and has the form 111acccccddjjj.

comp	$a\ c_1c_2c_3c_4c_5c_6$	dest	$d_1d_2d_3$	jump	$j_1j_2j_3$
D+M	1 000010	D	010	null	000

14. Using the generic machine-language notation used in class, give one example of each addressing mode and explain what is accessed: immediate, direct, and indirect addressing. [2 marks]
16. What is the role of (i) data registers, (ii) addressing registers, and (iii) the program counter in a CPU? [3 marks]
17. Answer the following questions regarding the Program Counter (PC). [4 marks]
- Give the specification of the PC in terms of its inputs, outputs, and control signals.
 - In Hack Computer architecture, where does the input wires of the PC come from?
 - Is the output of the PC connected to the RAM or the ROM? Give one or two line explanation.
18. The current values are PC=20 and A=50. Give the next value of the PC in each case. [3 marks]
- The instruction is D;JGT, the ALU output is 3, and **reset**=0.
 - The instruction is D;JGT, the ALU output is 0, and **reset**=0.
 - The instruction is 0;JMP and **reset**=1.

End of question paper